

APPENDIX II

Useful Relations in Cartesian and in Curvilinear Coordinates

1. CARTESIAN COORDINATES

Orthogonal unit vectors:

$$\hat{x}, \hat{y}, \hat{z}$$

Orthogonal line elements:

$$dx, dy, dz$$

Components of gradient:

$$(\vec{\nabla}\psi)_x = \frac{\partial\psi}{\partial x}$$

$$(\vec{\nabla}\psi)_y = \frac{\partial\psi}{\partial y}$$

$$(\vec{\nabla}\psi)_z = \frac{\partial\psi}{\partial z}$$

Divergence:

$$\vec{\nabla} \cdot \vec{A} = \frac{\partial}{\partial x} A_x + \frac{\partial}{\partial y} A_y + \frac{\partial}{\partial z} A_z$$

Components of curl:

$$(\vec{\nabla} \times \vec{A})_x = \left(\frac{\partial}{\partial y} A_z - \frac{\partial}{\partial z} A_y \right)$$

$$(\vec{\nabla} \times \vec{A})_y = \left(\frac{\partial}{\partial z} A_x - \frac{\partial}{\partial x} A_z \right)$$

$$(\vec{\nabla} \times \vec{A})_z = \left(\frac{\partial}{\partial x} A_y - \frac{\partial}{\partial y} A_x \right)$$

Laplacian:

$$\nabla^2 \psi = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2}$$

Components of divergence of a tensor:

$$(\vec{\nabla} \cdot \vec{\bar{T}})_x = \frac{\partial}{\partial x} T_{xx} + \frac{\partial}{\partial y} T_{yx} + \frac{\partial}{\partial z} T_{zx}$$

$$(\vec{\nabla} \cdot \vec{\bar{T}})_y = \frac{\partial}{\partial x} T_{xy} + \frac{\partial}{\partial y} T_{yy} + \frac{\partial}{\partial z} T_{zy}$$

$$(\vec{\nabla} \cdot \vec{\bar{T}})_z = \frac{\partial}{\partial x} T_{xz} + \frac{\partial}{\partial y} T_{yz} + \frac{\partial}{\partial z} T_{zz}$$

2. CYLINDRICAL COORDINATES

Orthogonal unit vectors:

$$\hat{\rho}, \hat{\phi}, \hat{z}$$

Orthogonal line elements:

$$d\rho, \rho d\phi, dz$$

Components of gradient:

$$(\vec{\nabla} \psi)_\rho = \frac{\partial \psi}{\partial \rho}$$

$$(\vec{\nabla} \psi)_\phi = \frac{1}{\rho} \frac{\partial \psi}{\partial \phi}$$

$$(\vec{\nabla} \psi)_z = \frac{\partial \psi}{\partial z}$$

Divergence:

$$\vec{\nabla} \cdot \vec{A} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial}{\partial \phi} A_\phi + \frac{\partial}{\partial z} A_z$$

Components of curl:

$$(\vec{\nabla} \times \vec{A})_\rho = \frac{1}{\rho} \frac{\partial}{\partial \phi} A_z - \frac{\partial}{\partial z} A_\phi$$

$$(\vec{\nabla} \times \vec{A})_\phi = \frac{\partial}{\partial z} A_\rho - \frac{\partial}{\partial \rho} A_z$$

$$(\vec{\nabla} \times \vec{A})_z = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\phi) - \frac{1}{\rho} \frac{\partial}{\partial \phi} A_\rho$$

Laplacian:

$$\nabla^2 \psi = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \psi}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 \psi}{\partial \phi^2} + \frac{\partial^2 \psi}{\partial z^2}$$

Components of divergence of a tensor:

$$(\vec{\nabla} \cdot \vec{\bar{T}})_\rho = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho T_{\rho\rho}) + \frac{1}{\rho} \frac{\partial}{\partial \phi} (T_{\phi\rho}) + \frac{\partial}{\partial z} T_{z\rho} - \frac{1}{\rho} T_{\phi\phi}$$

$$(\vec{\nabla} \cdot \vec{\bar{T}})_\phi = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho T_{\rho\phi}) + \frac{1}{\rho} \frac{\partial}{\partial \phi} T_{\phi\phi} + \frac{\partial}{\partial z} T_{z\phi} + \frac{1}{\rho} T_{\phi\rho}$$

$$(\vec{\nabla} \cdot \vec{\bar{T}})_z = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho T_{\rho z}) + \frac{1}{\rho} \frac{\partial}{\partial \phi} T_{\phi z} + \frac{\partial}{\partial z} T_{zz}$$

3. SPHERICAL COORDINATES

Orthogonal unit vectors:

$$\hat{r}, \hat{\theta}, \hat{\phi}$$

Orthogonal line elements:

$$dr, r d\theta, r \sin \theta d\phi$$

Components of gradient:

$$(\vec{\nabla}\psi)_r = \frac{\partial\psi}{\partial r}$$

$$(\vec{\nabla}\psi)_\theta = \frac{1}{r} \frac{\partial\psi}{\partial\theta}$$

$$(\vec{\nabla}\psi)_\phi = \frac{1}{r \sin \theta} \frac{\partial\psi}{\partial\phi}$$

Divergence:

$$\vec{\nabla} \cdot \vec{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} A_\phi$$

Components of curl:

$$(\vec{\nabla} \times \vec{A})_r = \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A_\phi \sin \theta) - \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} A_\theta$$

$$(\vec{\nabla} \times \vec{A})_\theta = \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} A_r - \frac{1}{r} \frac{\partial}{\partial r} (r A_\phi)$$

$$(\vec{\nabla} \times \vec{A})_\phi = \frac{1}{r} \frac{\partial}{\partial r} (r A_\theta) - \frac{1}{r} \frac{\partial}{\partial \theta} A_r$$

Laplacian:

$$\nabla^2 \psi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2}$$

Components of divergence of a tensor:

$$\begin{aligned}
 (\vec{\nabla} \cdot \vec{T})_r &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 T_{rr}) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (T_{\theta r} \sin \theta) + \\
 &+ \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} T_{\phi r} - \frac{1}{r} (T_{\theta\theta} + T_{\phi\phi})
 \end{aligned}$$

$$\begin{aligned}
 (\vec{\nabla} \cdot \vec{T})_\theta &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 T_{r\theta}) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (T_{\theta\theta} \sin \theta) + \\
 &+ \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} T_{\phi\theta} + \frac{1}{r} (T_{\theta r} - \cot \theta T_{\phi\phi})
 \end{aligned}$$

$$\begin{aligned}
 (\vec{\nabla} \cdot \vec{T})_\phi &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 T_{r\phi}) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (T_{\theta\phi} \sin \theta) + \\
 &+ \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} T_{\phi\phi} + \frac{1}{r} (T_{\phi r} + \cot \theta T_{\phi\theta})
 \end{aligned}$$